Capacitors

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Capacitance

A capacitor is used to store electrical energy in form of electrostatic field. This stored energy can then be utilised in a circuit as per the requirements.

A capacitor gets charged from a battery and this charge produces some energy inside it. This energy can be used in devices like a flashbulb, laser pulses, induction motors.

In thermonuclear fusion of hydrogen, short laser pulses are produced by very large capacitors.

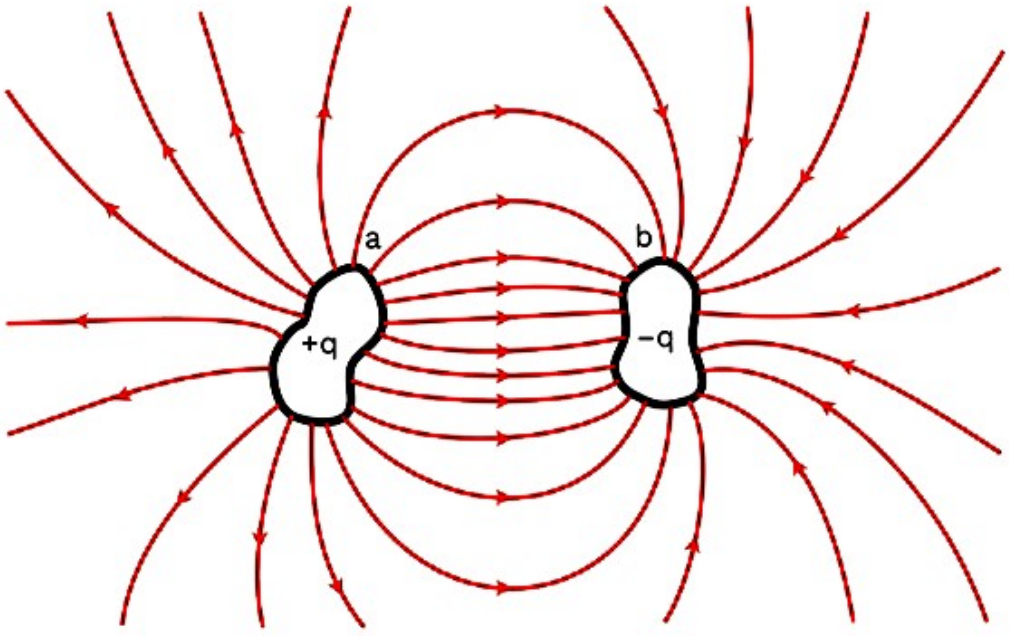
In oscilloscope tubes of older televisions, capacitors were used to produce electric fields which deflected the electron beams in the tubes to produce images on the screen.

Capacitors are used in tuning circuits which are used in radios, signal receivers in walkie-talkies, televisions etc.

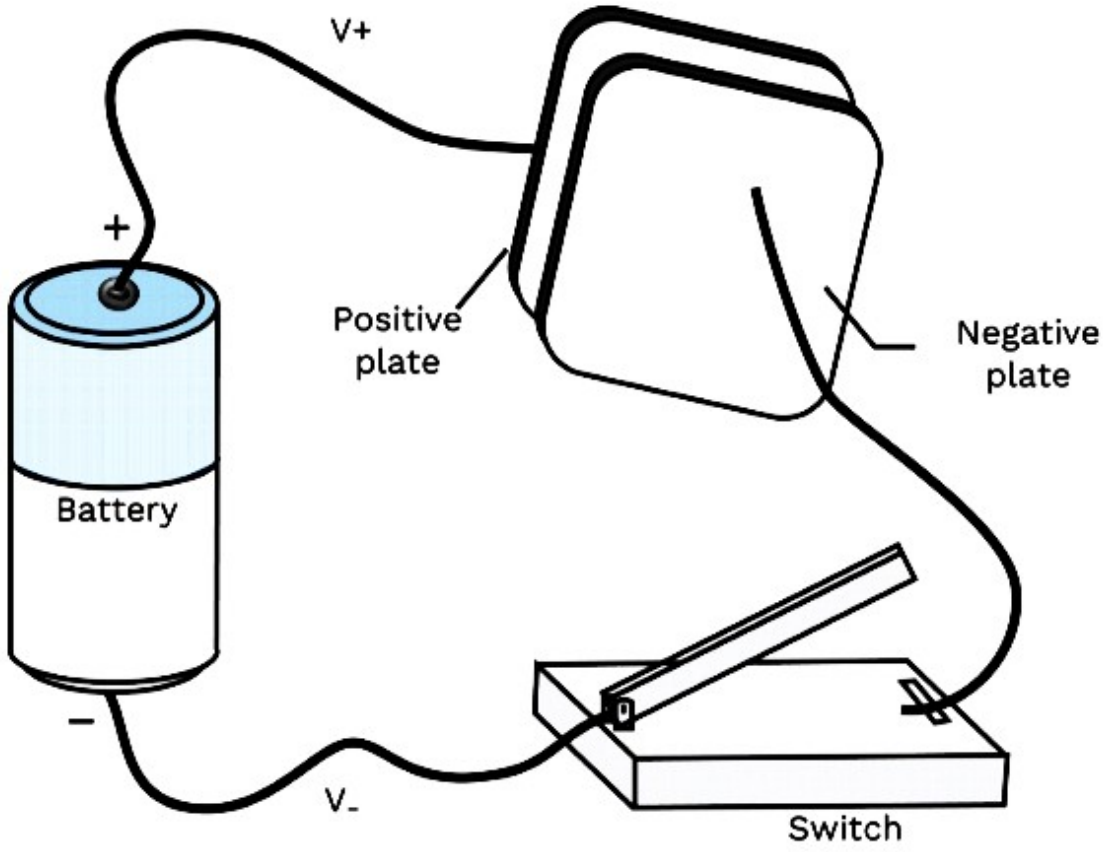
Capacitance

Two conducting bodies of whatever shape which are separated by a distance can form a capacitor. On giving equal and opposite charges to the bodies, the bodies can be termed as plates of the capacitor.

In the figure given below, we have two arbitrary shaped conductors kept in vacuum. The conductors are given charge +q and -q. So, net charge on the capacitor is zero, but magnitude of charge on each plate is q. We can see the electric field lines produced due to the charges on the bodies, and the electrical energy is stored in this electric field around the conducting bodies.



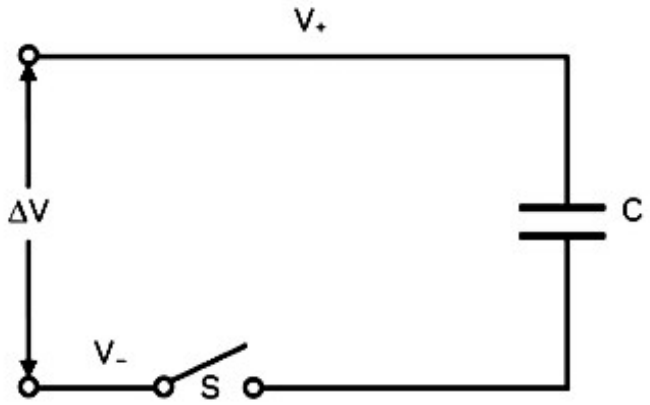
When we use two conducting plates kept parallel to each other and separated by a small distance, then we say that it is a parallel plate capacitor. A capacitor can be charged by connecting the plates to a battery as shown in the figure given below.



Battery is source of electrical energy, and it supplies charges to the connected plates. Positive charge is supplied to the plate connected to the positive terminal of the battery while negative charge is supplied to the battery connected to the negative terminal of the battery.

For an ideal battery connected to a capacitor, the potential of the positive plate of the capacitor and the potential of the positive terminal of the battery should be same i.e. V_+. Similarly, the potential of the negative terminal of the battery and the potential of the negative plate of the capacitor should be same i.e. V_-.

So, the potential difference between the terminals of the battery will be same as the potential difference between the plates of the capacitor.



The circuit symbol of capacitor is ⊣⊢. Two parallel lines separated by small distance represent the two plates of the capacitor.

While charging a capacitor, it is found that the charge on the plates of the capacitor is directly proportional to the potential difference between plates of the capacitor.

$q \propto \Delta V$.

We can remove this proportionality sign as,

$q = C\,\Delta V$.

where C is the capacitance of the capacitor formed.

SI unit of capacitance is farad (F) which is equal to 1 coulomb per volt. The unit farad is used for capacitance in honour of its inventor Michael Faraday.

Smaller units for capacitance such as microfarad (1 μF = 10^{-6} F) and picofarad (1 pF = 10^{-12} F) are more generally used.

Capacitance only depends on the shape, size, separation between the conductors of the capacitor and the medium in between the plates and is independent of the charge or the potential difference across the capacitor.

Q. **A storage capacitor on a random-access memory (RAM) chip has a capacitance of 0.055 pF. If this storage capacitor is charged to 5.3 V, then what is the number of excess electrons on its negative plate??**

Sol. For a charge q on the negative plate of the capacitor, the number of excess electrons, $N = \frac{q}{e} = \frac{C\Delta V}{e} = \frac{(0.055 \times 10^{-12}\,F)(5.3V)}{1.60 \times 10^{-19}C} = 1.8 \times 10^{6}$ electrons.

Note that the result obtained is a very small number when we talk about the number of electrons. Very small dust particles in air have about 10^{17} electrons.

Calculating The Capacitance

To find the capacitance of a capacitor, the following procedure is followed:

1. Electric field between the plates of the capacitor is calculated first.

2. Then, a suitable path is chosen between the positive and negative plates of the capacitor and the potential difference is calculated by integrating the expression dV=-E.ds along the chosen path.

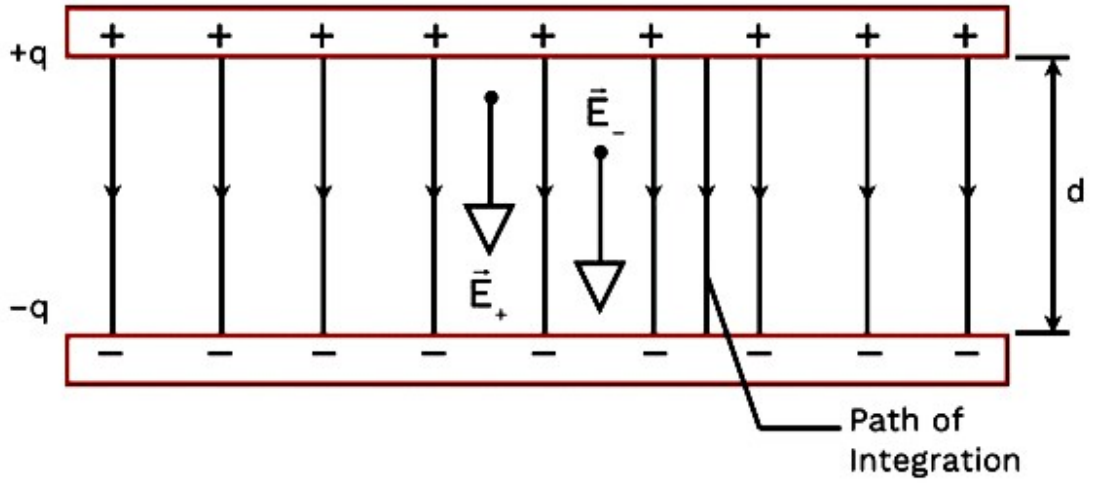
3. We then use the relation, $C = \frac{q}{\Delta V}$ to find the capacitance of the given arrangement of the conductors.

Capacitance of a system of conductors is always a positive quantity. The above procedure is now applied on different arrangements of conductors and their capacitances are calculated below.

A Parallel-Plate Capacitor

Two large parallel plates separated by a small distance form a parallel plate capacitor. The separation between the plates should be very less as compared to the length and width of the plates. If +q is the charge on the positive plate and -q is the charge on the negative plate of the capacitor, then surface charge density on the plates will be.

$\sigma = \frac{q}{A}$, where A is the area of the plate.



The electric field due to one charged plate is given as,

$$E = \frac{\sigma}{2\varepsilon_0}.$$

We are using the result of an infinite charged plate here due to the reason that the space between the plates is very small as compared to the plate dimensions.

An electric field of same magnitude will be present due to the negatively charged plate in the same direction as that of the field due to the positively charged plate.

Total electric field between the plates : $\vec{E} = \vec{E}_+ + \vec{E}_-$.

$$E = E_+ + E_- = \frac{\sigma}{2\varepsilon_0} + \frac{\sigma}{2\varepsilon_0} = \frac{\sigma}{\varepsilon_0}$$

Put $\sigma = \frac{q}{A}$ to get,

$$\Delta V = \int_+^- E ds = \frac{q}{\varepsilon_0 A}\int_+^- ds = \frac{qd}{\varepsilon_0 A},$$

The integration path is chosen along one of the electric field lines between the plates.

On comparing above result with the standard formula $C = q/\Delta V$, we get

$C = \frac{\varepsilon_0 A}{d}$ (which gives the capacitance of a Parallel-plate capacitor).

The above result clearly shows that the capacitance depends on geometrical factors only, which in this case are the plate separation d and area A. The capacitance is independent of the potential difference between the plates, or the charge on the plates.

Also, from the expression of capacitance obtained above, we can see that the unit of ε_0 can be written as the capacitance divided by length, i.e., the unit becomes F/m.

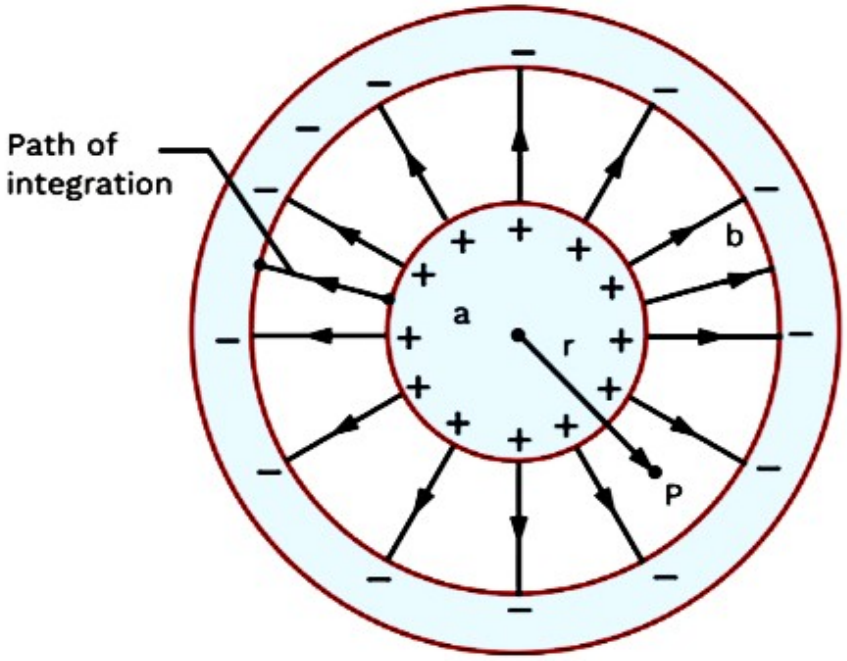
$\varepsilon_0 = 8.85 \times 10^{-12}$ F/m = 8.85 pF/m.

This unit for ε_0 is generally more used as compared to the unit $C^2/N\ m^2$ obtained from Coulomb's law.

A Spherical Capacitor

Consider a solid sphere of radius "a" surrounded by a hollow spherical shell of radius "b" as shown in the figure below.

If a uniformly distributed charge q is given to the inner sphere, then there will be induced charge -q on the inner surface of the spherical shell. The electric field line originating from the positive charge on the inner sphere will terminate at the negative charges on the inner surface of the hollow sphere. Thus, we get a capacitor arrangement here.



Now, we can use Gauss law to find the value of electric field in the region between the inner sphere and the shell.
Draw a gaussian surface of radius r such that a< r < b. Then the electric field at radial distance r will be,

$$E = \frac{1}{4\pi\varepsilon_0}\frac{q}{r^2},$$

Now, we can choose the radial path for integration in order to find the potential difference between the surface of the sphere and the inner surface of the shell as,

$$\Delta V = \int_{\cdot}^{-} E ds = \int_{a}^{b} \frac{q}{4\pi\varepsilon_0}\frac{dr}{r^2} = \frac{q}{4\pi\varepsilon_0}\left(\frac{1}{a} - \frac{1}{b}\right).$$

$$= \frac{q}{4\pi\varepsilon_0}\left(\frac{b-a}{ab}\right)$$

As the integration path is radial, so $\vec{E}\cdot\vec{ds} = Eds$ and ds = dr.
Using C = q/ΔV, we can write,

$$C = 4\pi\varepsilon_0\left(\frac{ab}{b-a}\right)$$

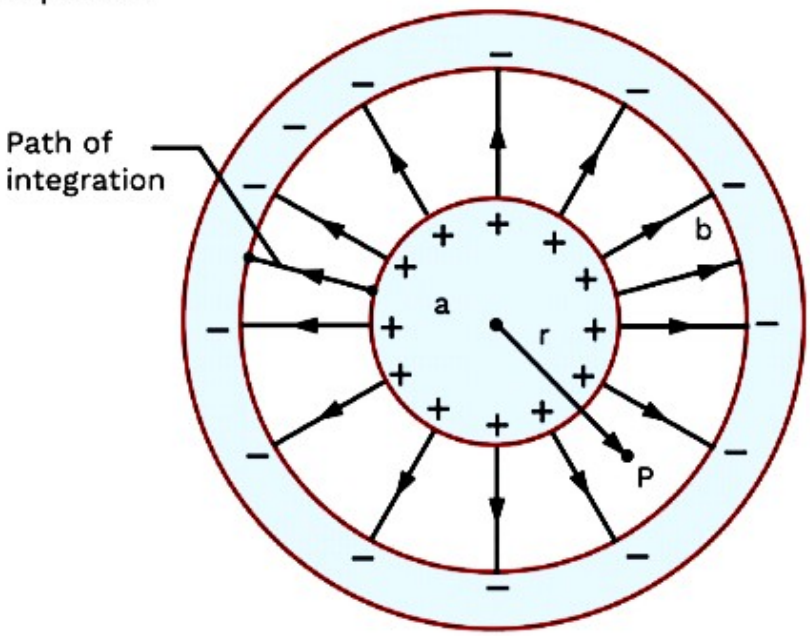
Again, from above relation, we can see that the capacitance of the spherical capacitor is just depending on the geometrical factors of the capacitor.

A Cylindrical Capacitor

Consider a solid cylinder of radius "a" which is surrounded by a hollow cylinder of radius "b". The figure of cross section of the considered cylindrical system is shown in the figure below. If a charge q is given to the inner solid cylinder, then there will be induced negative charge on the inner surface of the hollow cylinder.

The electric field lines originate radially from the positive charge on the inner cylinder and terminate at the negative charge on the inner surface of the outer cylinder.

Thus, we get a capacitor arrangement. Assume the cylinder to be of length L and L to be very greater than the radius "b", we can neglect the leakage of electric field lines (fringing effect) at the ends of the capacitor.



Now, again by using Gauss law, we can find the value of electric field in the region between the solid cylinder and the inner surface of the outer cylinder.

Draw a cylindrical gaussian surface of radius r such that a < r < b. then by Gauss law, we can find the electric field as,

$$E = \frac{1}{2\pi\varepsilon_0}\left(\frac{\lambda}{r}\right) = \frac{1}{2\pi\varepsilon_0}\left(\frac{q}{Lr}\right)$$

Choosing the integration path to be radial, we can write,

$$\Delta V = \int_{+}^{-} Eds = \frac{q}{2\pi\varepsilon_0 L}\int_a^b \frac{dr}{r} = \frac{q}{2\pi\epsilon_0 L}\ln\left(\frac{b}{a}\right).$$

On comparing with the standard equation $C = q/\Delta V$, we get,

$$C = 2\pi\varepsilon_0 \frac{L}{\ln\left(\frac{b}{a}\right)}$$

Again, the capacitance is only depending upon the geometrical factors.

Q. **The plates of a parallel-plate capacitor are separated by a distance d = 1.0 mm. What must be the plate area if the capacitance is to be 1.0 F?**

Sol. From the formula of capacitance of parallel plate capacitor,

$$A = \frac{Cd}{\varepsilon_0} = \frac{(1.0F)(1.0\times10^{-3}m)}{8.85\times10^{-12}F/m} = 1.1\times10^{8}m^2$$

Note that the value of the area obtained is very large. If we take a square plate with side length of about 10 km, then we get this value of the area. This shows that 1 F is a very large unit for capacitance. Generally used capacitors are of micro-farad, nano-farad or pico-farad capacities.

Q. **The space between the conductors of a long coaxial cable, used to transmit TV signals, has an inner radius a = 0.15 mm and an outer radius b = 2.1 mm. What is the capacitance per unit length of this cable?**

Sol. From equation of capacitance for cylindrical capacitor, we have

$$\frac{C}{L} = \frac{2\pi\varepsilon_0}{\ln(b/a)} = \frac{(2\pi)(8.85pF/m)}{\ln(2.1mm/0.15mm)} = 21\ pF/m$$

Q. **What is the capacitance of the Earth, viewed as an isolated conducting sphere of radius R = 6370 km?**

Sol. An isolated solid conducting sphere will also act as a capacitor. The outer shell will be considered as a sphere of infinite radius.

Then, in the formula of the capacitance for spherical capacitor,

$b \to \infty$ and a = R, Then,

$C = 4\pi\varepsilon_0 R$ (isolated sphere).

Substituting the values, we obtain

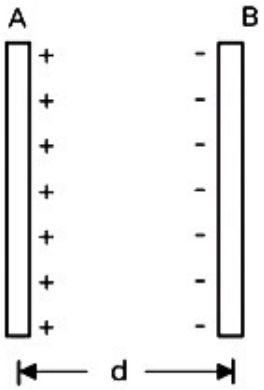
$C = (4\pi)\ (8.85 \times 10^{-12}\ F/m)\ (6.37 \times 10^{6}\ m)$

$= 7.1 \times 10^{-4}\ F = 710\ \mu F.$

Force Between The Plates Of Capacitor :-

Electric field between the plates of capacitor

$$E = \frac{\sigma}{\epsilon_0}$$



Electric force on B due to charge on A,

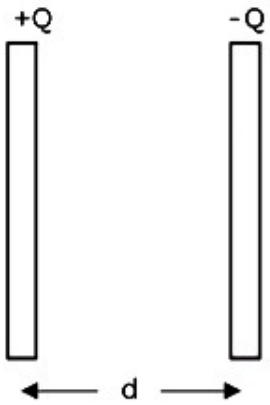
$$F = q_B \times E_A$$

$$= Q\left(\frac{\sigma}{2\,\epsilon_0}\right)$$

$$F = \frac{Q^2}{2A\,\epsilon_0} \qquad \left(\sigma = \frac{Q}{A}\right)$$

Energy stored in a capacitor :

As we know that



$$F = -\frac{dU}{dr}$$

$$dU = -F\,dr$$

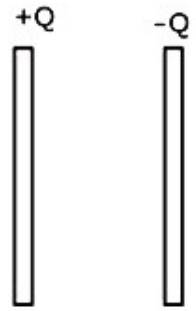
$$U = -\frac{Q^2}{2A\,\epsilon_0}\int_0^d dr$$

$$= \frac{Q^2 d}{2A\,\epsilon_0} = \frac{Q^2}{2A\,\epsilon_0\,/d} = \frac{Q^2}{2C}$$

Energy Density :

There are two capacitor plates with charges +Q and -Q, then,

Energy, $U = \frac{Q^2}{2C} = \frac{\sigma^2 A^2 d}{2A\,\epsilon_0}$



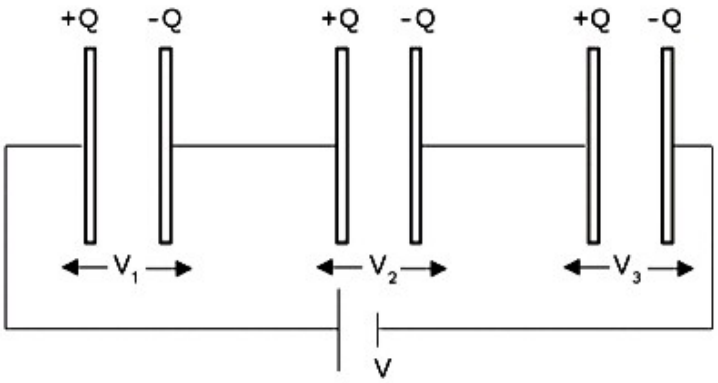
$$= \frac{\sigma^2}{\epsilon_0^2}\frac{Ad\,\epsilon_0}{2}$$

Energy Density $= \frac{U}{Ad} = \frac{\epsilon_0 E^2}{2}$

(Energy stored /vol), $U = \frac{\epsilon_0 E^2}{2}$

Combination of Capacitors

Capacitors in series :-



In the series combination of capacitors, all capacitors store same amount of charge.

$V = V_1 + V_2 + V_3$

$$\frac{Q}{C_{net}} = \frac{Q}{C_1} + \frac{Q}{C_2} + \frac{Q}{C_3}$$

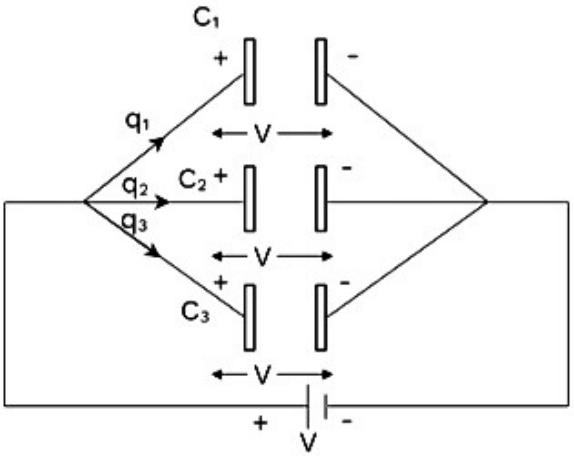
$$\frac{1}{C_{net}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} + ...$$

Now these are two main points for series combination :-

(1) Each capacitor will get same amount of charge.

(2) Potential difference across each capacitor is inversely proportional to its capacitance.

Parallel Combination of Capacitors :-



In parallel combination, the potential difference across each capacitor is same, as that of the external battery.
But charge will be divided which is proportional to their capacitance.

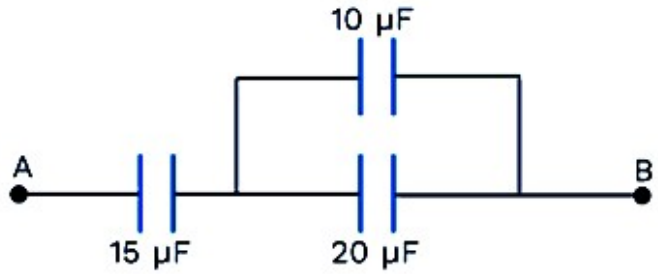
$Q_1 = C_1V \qquad Q_2 = C_2V \qquad Q_3 = C_3V$

$Q = Q_1 + Q_2 + Q_3$

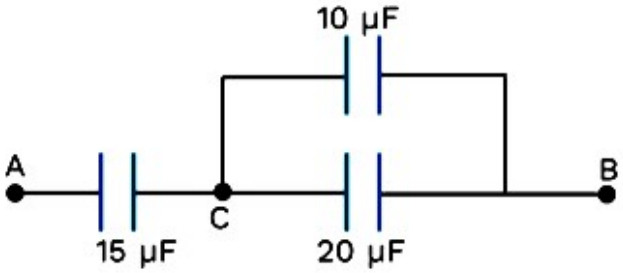
$C_{net}V = C_1V + C_2V + C_3V$

$C_{net} = C_1 + C_2 + C_3$

Q. **What is the equivalent capacitance between A and B in the figure shown below?**



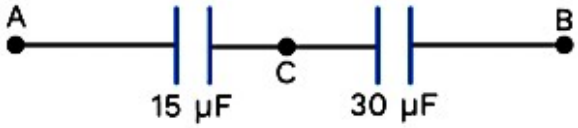
Sol. First, mark different junctions (nodes) in the circuit. A node is a point where more than two wires are connected.
Capacitors 20 μF and 10 μF are connected between same points C and B (so, it is a Parallel combination)



Their parallel equivalent will be,

$C = 10\ \mu F + 20\ \mu F = 30\ \mu F$

Thus, the circuit can be shown like this.



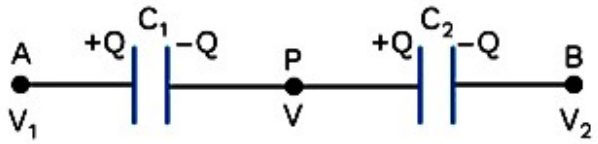
Now, the capacitors 15 μF and 30 μF are in series combination.
∴ The equivalent capacitance is given by C such that,

$$\frac{1}{C} = \frac{1}{15} + \frac{1}{30} \text{ or } C = 10\ \mu C$$

Q. **The arrangement shows a part of complete circuit. The potentials at points A and B are V_1 and V_2 respectively. Find the potential at point P.**



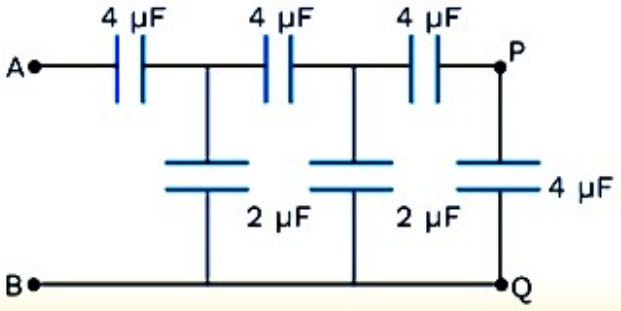
Sol. The two capacitors C_1 and C_2 are connected in series. Therefore, the charge on both capacitors is same. Let the potential at P is V.



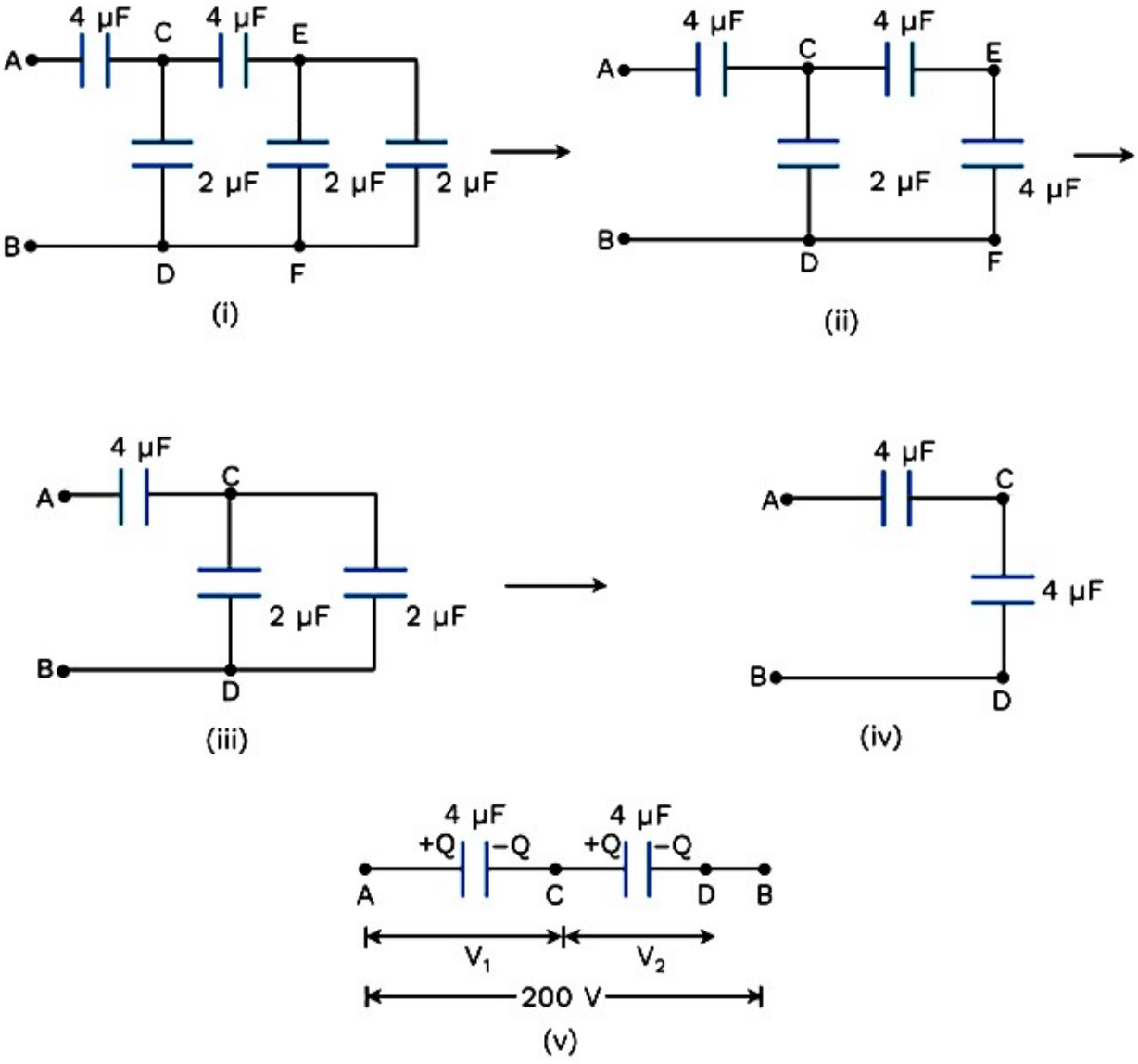
∴ Potential difference across C_1 is $V_1 - V$ and across C_2 is $V - V_2$.
Let charge on C_1 is Q, such that, $Q = C_1(V_1 - V)$
Also, charge on C_2 is Q, such that, $Q = C_2(V - V_2)$
$\Rightarrow$ $C_1(V_1 - V) = C_2(V - V_2)$

$$\Rightarrow \quad V = \frac{C_1V_1 + C_2V_2}{C_1 + C_2}$$

Q. **In the arrangement shown, a potential difference of 200 V is set up between A and B. What is the potential difference between P and Q as shown?**



Sol. First let us find the equivalent capacitance between A and B. Various steps of circuit simplification involved are as shown. Different junctions have been marked.



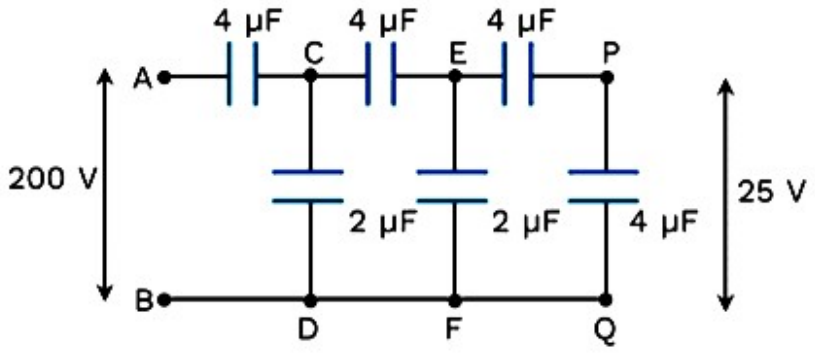
$\Rightarrow$ $C_{AB} = 2\ \mu F$

$\Rightarrow$ $Q = CV = 400\ \mu C$

Potential difference, $V_C - V_D = V_2 = \frac{Q}{C} = \frac{400}{4} = 100\,V$ (as, $C_1 = C_2$, $V_1 = V_2 = 100$ V)

$\Rightarrow$ Potential difference, $V_E - V_F = 50$ V.

$\therefore$ Potential difference $V_P - V_Q = 25$ V.



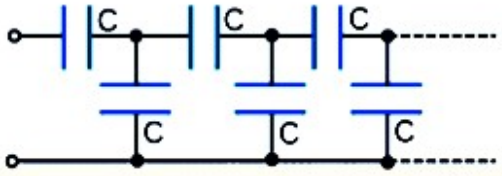
Q. **Consider we have two capacitors C_1 = 1.0 μF and C_2 = 2.0 μF. The maximum voltages that they can withstand are V_1 = 6.0 kV and V_2 = 4.0 kV respectively. For a series combination of these two capacitors, what os the maximum voltage that can be applied?**

Sol. Maximum charge that can be on the capacitor of capacitance C_1, $q_1 = C_1V_1$, and,
Maximum charge on the capacitor of capacitance C_2, $q_2 = C_2V_2$.
In the series combination, charge on both capacitors will be same.
So, maximum charge that the series combination can withstand = C_1V_1,
(...as $C_1V_1 < C_2V_2$, from the numerical data, given.)
Now, net capacitance of the system,

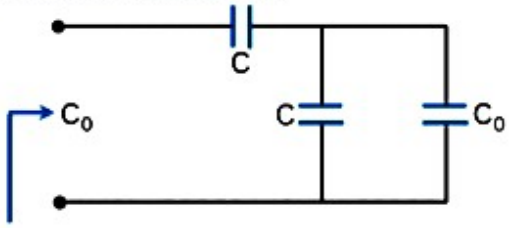
$$C_0 = \frac{C_1C_2}{C_1 + C_2}$$

and hence, $V_{max} = \frac{q_{max}}{C_0} = \frac{C_1V_1}{\left(\frac{C_1C_2}{C_1 + C_2}\right)} = V_1\left(1 + \frac{C_1}{C_2}\right) = 9kV$

Q. **What is the capacitance of an infinite circuit formed by the repetition of the same link having two identical capacitors, each with capacitance C?**



Sol. The equivalent capacitance of the arrangement does not depend on the number of branches. So, if we add or subtract one more branch in the network, the equivalent capacitance remains the same.
The infinite circuit, may be reduced to the circuit, shown in the fig where C_0 is the net capacitance of the combination.

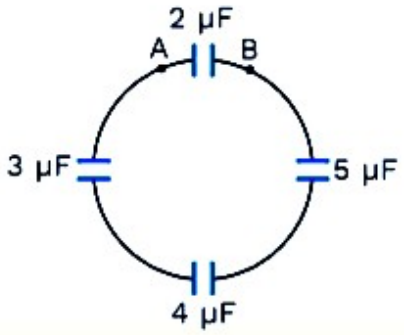


So, $\frac{1}{C + C_0} + \frac{1}{C} = \frac{1}{C_0}$

Solving the quadratic equation, $CC_0 + C_0^2 - C^2 = 0$,

we get, $C_0 = \frac{\left(\sqrt{5} - 1\right)}{2} C$, taking only +ve values as C_0 cannot be negative

Q. **For the following arrangement, find the equivalent capacitance between A and B.**



Sol. Here, the 3 μF, 4 μF and 5 μF capacitors are in series

$$\therefore \quad C_{eq} = \left(\frac{1}{3} + \frac{1}{4} + \frac{1}{5}\right)^{-1} = \frac{60}{12 + 12 + 15} = \frac{60}{47}\mu F$$

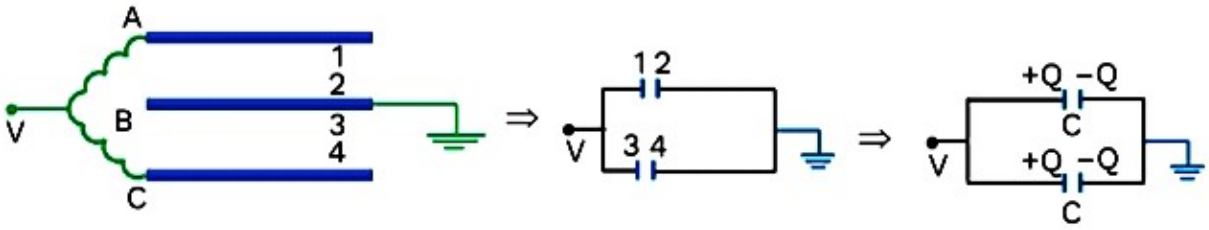
Now, 2 μF and $\frac{60}{47}\mu F$ are in parallel, so we have

$$C_{eq} = C_1 + C_2 = \frac{154}{47}\mu F$$

Q. **Find the charge on each plate, given area of each plate is A and separation between the consecutive plates is d.**



Sol. Let the plates are A, B and C. This an arrangement of two capacitors. First, we identify how the capacitors are connected.



The faces 1 & 2 form a capacitor out of which, 1 is connected to V volts and 2 is grounded. Similarly, faces 3 and 4 form a capacitor out of which, 3 is grounded and 4 is connected to V volts. The arrangement is two capacitors in parallel.

So, we have $Q = CV = \left(\frac{\varepsilon_0 A}{d}\right)V$

So, net charge on plate A is $\left(\frac{\varepsilon_0 A}{d}\right)V$

Net charge on plate B is equal to sum of charges on faces 2 and 3, which is

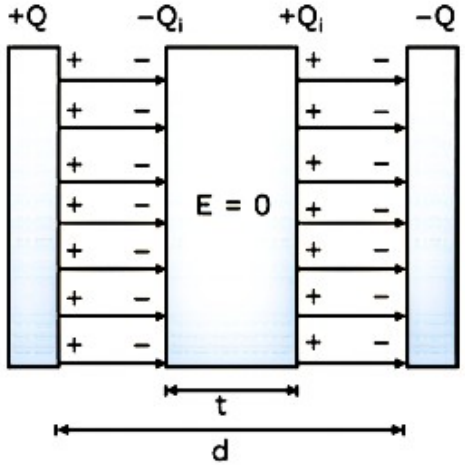
equal to $-\left(\frac{2\varepsilon_0 A}{d}\right)V$.

Net charge on plate C is $\left(\frac{\varepsilon_0 A}{d}\right)V$

Capacitors with Dielectrics

Effect of a Metallic Siab in a Parallel Plate Capacitor

Consider a parallel plate capacitor with plate separation d and plate area A. The surface charge density is σ. Electric field in the free space is $E_0 = \frac{\sigma}{\varepsilon_0}$ and inside the slab, E = 0. This is so because the induced electric field balances the applied electric field.



External field, $E_0 = \frac{\sigma}{\varepsilon_0}$

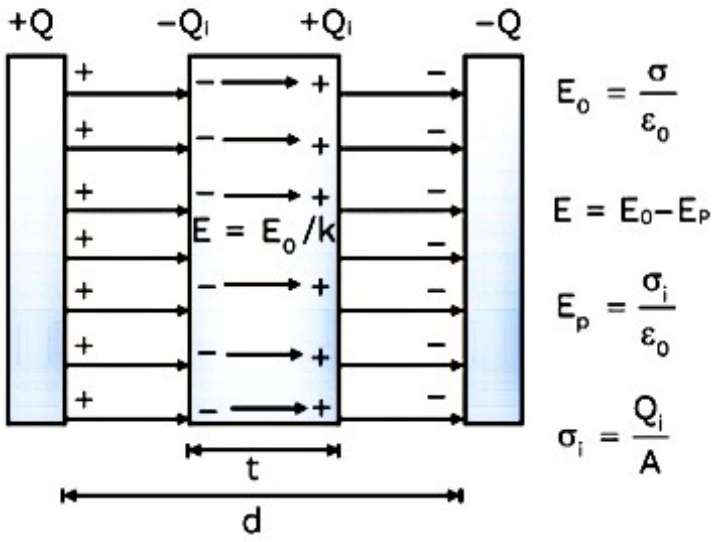
Induced field, $E_i = \frac{\sigma_i}{\varepsilon_0}$
Plate area = A
AS $E_i - E_0$ = Net field = 0 $\Rightarrow E_i = E_0$
$\Rightarrow \sigma_1 = \sigma$
$\Rightarrow Q_i = Q$
The potential difference between the plates can be calculated as

$$V = E_0(d - t) = \frac{\sigma}{\varepsilon_0}(d - t) = \frac{Q}{A\varepsilon_0}(d - t)$$

As, $C = \frac{Q}{V} \Rightarrow C = \frac{\varepsilon_0 A}{d - t}$ (i.e., Capacitance increases)

Effect of Dielectric Slab (Inserted along the length of plates)

When a dielectric slab is placed between the plates of capacitor, its polarisation takes place. Thus, a charge $-Q_i$, appears on its left face and $+Q_i$, appears on its right face, as shown in diagram.



This causes a decrease in electric field between the plates.

The net field inside a dielectric is E, such that $E = \frac{E_0}{K}$

Here K is dielectric constant (or ε_r that is relative permittivity)
Net field is also given $E = E_0 - E_p$ (where E_p is electric field due to polarisation)

$$\therefore \quad \frac{E_0}{K} = E_0 - E_P \qquad \Rightarrow \qquad E_p = E_0 - \frac{E_0}{K}$$

$$\frac{\sigma_i}{\varepsilon_0} = \frac{\sigma}{\varepsilon_0} - \frac{\sigma}{K\varepsilon_0} \qquad \Rightarrow \qquad Q_i = Q\left(1 - \frac{1}{K}\right)$$

Potential difference between the plates can be calculated as,

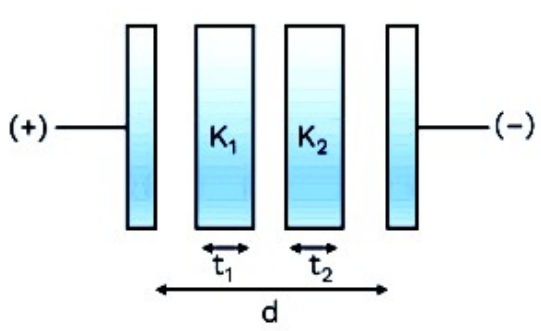
$$V = E_0(d - t) + Et = E_0\left(d - t + \frac{t}{K}\right)$$

$$\therefore \quad V = \frac{\sigma}{\varepsilon_0}\left(d - t + \frac{t}{K}\right) = \frac{Q}{A\varepsilon_0}\left(d - t + \frac{t}{K}\right)$$

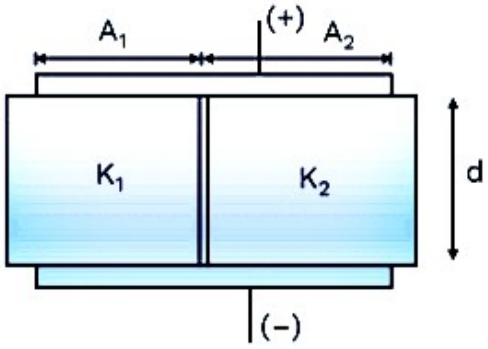
$$\text{As, } C = \frac{Q}{V} \qquad \Rightarrow \qquad C = \frac{\varepsilon_0 A}{d - t + \frac{t}{K}}$$

Note: When t = d, $C = \frac{K\varepsilon_0 A}{d}$

Q. **An air-cored capacitor of plate area A and separation d has a capacity C. Two dielectric slats are inserted between its plates in two different manners as shown. Calculate the capacitance in each case.**



(i)



(ii)

Sol. (i) Let the charges on the plates are Q and −Q.

Electric field in free space is $E_0 = \frac{\sigma}{\varepsilon_0} = \frac{Q}{A\varepsilon_0}$.

Electric field in first slab is $E_1 = \frac{E_0}{K_1} = \frac{Q}{A\varepsilon_0 K_1}$.

Electric field in second slab is $E_2 = \frac{E_0}{K_2} = \frac{Q}{A\varepsilon_0 K_2}$.

The potential difference between the plates is $V = E_0 (d - t_1 - t_2) + E_1 t_1 + E_2 t_2$.

$$\Rightarrow \quad V = E_0 \left(d - t_1 - t_2 + \frac{t_1}{K_1} + \frac{t_2}{K_2} \right) \qquad \left(\text{As } E_1 = \frac{E_0}{K_1}, E_2 = \frac{E_0}{K_2} \right)$$

$$\therefore \quad V = \frac{Q}{A\varepsilon_0} \left(d - t_1 - t_2 + \frac{t_1}{K_1} + \frac{t_2}{K_2} \right)$$

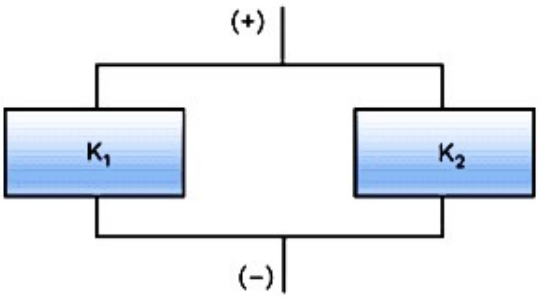
$$\therefore \quad C = \frac{\varepsilon_0 A}{d - t_1 - t_2 + \frac{t_1}{K_1} + \frac{t_2}{K_2}}$$

(ii) The capacitor can be considered as two capacitors of capacitances C_1 and C_2 in parallel

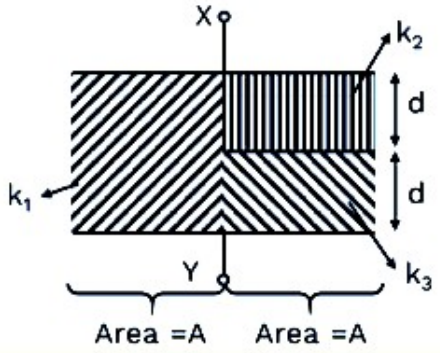
$$\Rightarrow \quad C_1 = K_1\frac{\varepsilon_0 A_1}{d}, \quad C_2 = \frac{K_2 + \varepsilon_0 A_2}{d}$$

$$\Rightarrow \quad \text{Total capacitance } C = C_1 + C_2$$

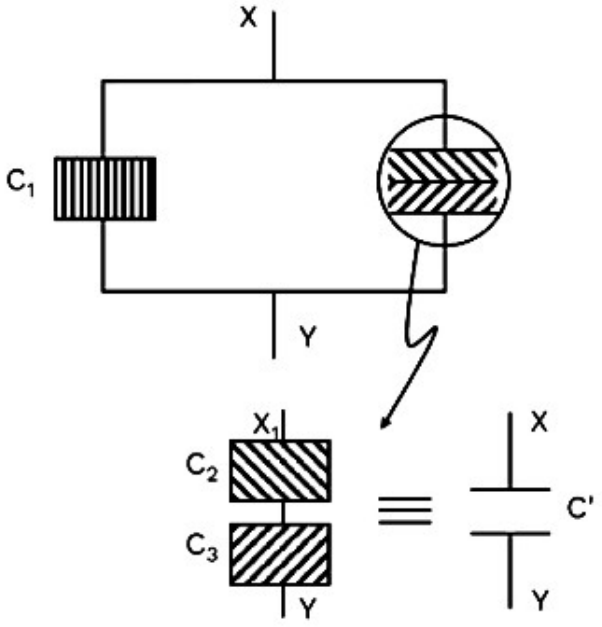
$$\Rightarrow \quad C = \frac{(K_1A_1 + K_2A_2)\varepsilon_0}{d}$$



Q. **Calculate the capacitance of the parallel plate capacitor shown in the following figure, between points X and Y.**



Sol.



The capacitor can be broken apart into the following circuit.

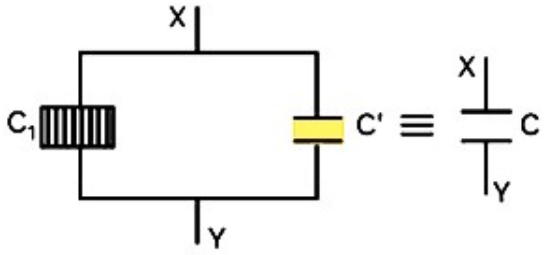
(i) Series (C_2 and C_3)

$$C_2 = \varepsilon_0 k_2 \frac{A}{d}, \quad C_3 = \varepsilon_0 k_3 \frac{A}{d}$$

$$\Rightarrow \frac{1}{C'} = \frac{1}{C_2} + \frac{1}{C_3}$$

(ii) Parallel ($C_1 \parallel C'$)

$$C_1 = \varepsilon_0 k_1 \frac{A}{2d}.$$



⇒Equivalent capacitance between X and Y is

$$C = C_1 + C' = C_1 + \frac{C_2 C_3}{C_2 + C_3}$$

So, the capacitance is $\left(\frac{k_1k_2 + 2k_2k_3 + k_3k_1}{2(k_2 + k_3)}\right)\frac{\varepsilon_0 A}{d}$

Q. An air-cored parallel plate capacitor having a capacitance C is charged by connecting it to a battery. Now, it is disconnected from the battery and a dielectric slab of dielectric constant K is inserted between its plates so as to completely fill the space between the plates. Compare:

(i) Initial and final capacitance.

(ii) Initial and final charge.

(iii) Initial and final potential difference.

(iv) Initial and final electric field between the plates.

(v) Initial and final energy stored in the capacitor.

Sol. (i) Initial capacitance = C
Final capacitance = KC (after inserting dielectric)

$$\Rightarrow \quad \frac{C_f}{C_i} = K$$

(ii) Charge on isolated capacitor remains same, therefore $\frac{Q_f}{Q_i} = 1$

(iii) $As,\ Q = CV \Rightarrow V = \frac{Q}{C} \quad \Rightarrow \frac{V_f}{V_i} = \frac{C_i}{C_f} = \frac{1}{K}$

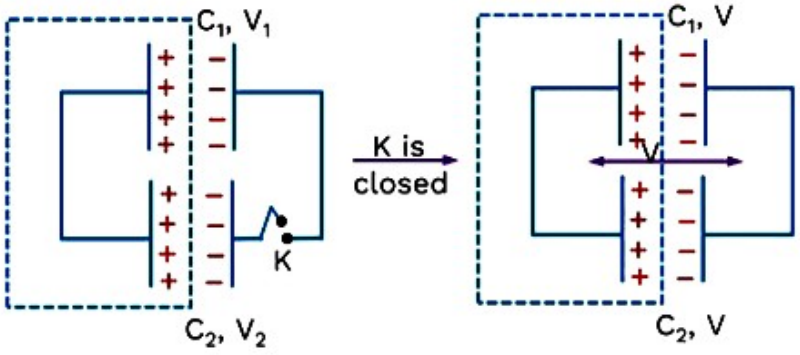
(iv) $E = \frac{V}{d} \quad \Rightarrow \frac{E_f}{E_i} = \frac{V_f}{V_i} = \frac{1}{K}$

(v) $U = \frac{Q^2}{2C} \quad \Rightarrow \frac{U_f}{U_i} = \frac{C_i}{C_f} = \frac{1}{K}$

Sharing of Charge and Common Potential

When two isolated capacitors (either both of them are charged or one of them is charged) are connected to each other, redistribution of charge takes place due to potential difference between them. Charge flows from higher potential to lower potential till both of them acquire same potential. This is called common potential.

Case-1:
Two capacitors, C_1 and C_2 charged to potentials, V_1 and V_2, are connected as shown, so that their similar plates are connected to each other.



Consider the part of circuit inside dotted loop. This part is isolated from other part. Therefore, total charge of this part remains constant. Hence,

$$C_1V_1 + C_2V_2 = C_1V + C_2V$$

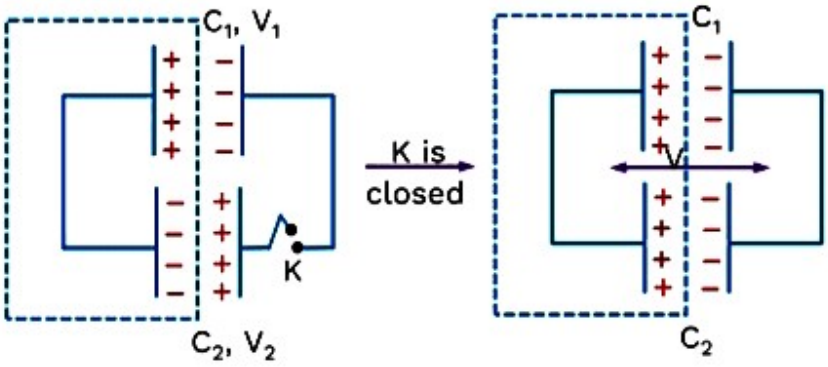
or, $$V = \frac{C_1V_1 + C_2V_2}{C_1 + C_2}$$

During this redistribution of charge, there is a loss of energy in the form of heat and electromagnetic radiations. Loss of energy, $\Delta U = U_i - U_f$

$$\Rightarrow \Delta U = \frac{1}{2}C_1V_1^2 + \frac{1}{2}C_2V_2^2 - \frac{1}{2}(C_1 + C_2)V^2$$

$$= \frac{1}{2}\left(\frac{C_1C_2}{C_1 + C_2}\right)(V_1 - V_2)^2 \qquad \left[\text{as } V = \frac{C_1V_1 + C_2V_2}{C_1 + C_2}\right]$$

Case-2:
The two capacitors are connected in manner, so that their dissimilar plates are connected together.



Again following the conservation of charge,

$$C_1V_1 - C_2V_2 = C_1V + C_2V$$

or, $$V = \frac{C_1V_1 - C_2V_2}{C_1 + C_2}$$

As calculated earlier, loss of energy can be calculated equal to, $\Delta U = \frac{1}{2}\frac{C_1C_2}{C_1 + C_2}(V_1 + V_2)^2$.

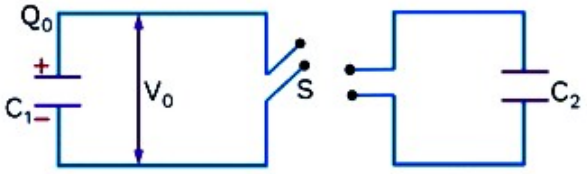
Q. **A parallel plate capacitor of capacitance C has been charged, so that the potential difference between its plates is V. Now, the plates of this capacitor are connected to another uncharged capacitor of capacitance 2C. Find the common potential acquired by the system and loss of energy**

Sol. Here $C_1 = C$, $C_2 = 2C$, $V_1 = V$ and $V_2 = 0$

$$\Rightarrow \quad \text{Common potential} = V = \frac{C_1V_1 + C_2V_2}{C_1 + C_2} = \frac{CV + 0}{3C} = \frac{V}{3}$$

And, $$\Delta U = \frac{1}{2}\frac{C \times 2C}{3C} \times (V - 0)^2 = \frac{1}{3}Cv^2$$

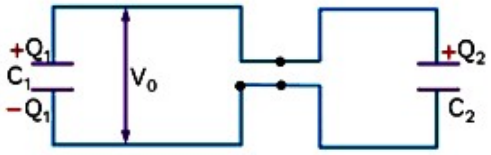
Q. **In the circuit shown, $C_1 = 8.0\ \mu F$, $C_2 = 4.0\ \mu F$ and $V_0 = 120V$.**



When the switch S is closed, the charged capacitor C_1 is connected to an uncharged capacitor C_2.
(a) What is the charge on each capacitor after switch S is closed?
(b) What is total energy after switch S is closed?

Sol. (a) Before switching on S, $Q_0 = C_1V_0 = 960\ \mu C$

Energy stored, $U = \frac{1}{2}Q_0V_0 = \frac{1}{2}(960 \times 10^{-6}\,C)(120\,V) = 0.058\ J$



(b) After we close switch S, $Q_0 = Q_1 + Q_2$
In the steady state,

$Q_1 = C_1V$ ...(1)
$Q_2 = C_2V$ ...(2)

Adding Equation (1) & (2),

$(C_1 + C_2)V = Q_1 + Q_2 = Q_0$

or $$V = \frac{Q_0}{C_1 + C_2} = \frac{960\,\mu C}{12\,\mu F} = 80V$$

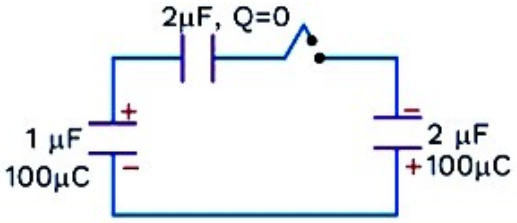
$\Rightarrow$ $Q_1 = 640\ \mu C$, $Q_2 = 320\ \mu C$
The total energy stored is given by

$$U = \frac{1}{2}Q_1V + \frac{1}{2}Q_2V = \frac{1}{2}Q_0V$$

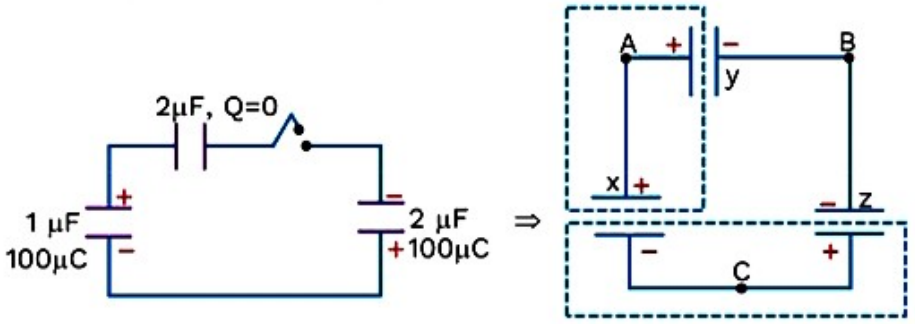
$$= \frac{1}{2}(960 \times 10^{-6})(80\,V) = 0.038\ J$$

The energy is less than the original energy of 0.058 J, the difference has been converted to energy of some other form (e.g. radiated as electromagnetic waves).

Q. **In the circuit shown, the capacitance of each capacitor and the charge on it are marked. What will be the charge on each capacitor, after the switch is closed?**



Sol. Let after closing the switch, charges are x, y and z respectively (in μC)



By conservation of charge in the isolated parts of the circuit

$x + y = 100$ ...(1)

$-x + z = 0$ or, $x = z$ ...(2)

Also, by $V = \frac{Q}{C}$, $V_A - V_B = \frac{y}{2}$

$$V_B - V_C = \frac{-z}{2}$$

$$V_C - V_A = \frac{-x}{1}$$

Adding these three equations,

$$0 = \frac{y}{2} - \frac{z}{2} - \frac{x}{1} \text{ or } 2x + z - y = 0 \quad ...(3)$$

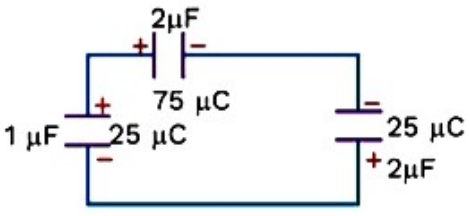
From (1), (2) and (3),

$z = 25\ \mu C$

$x = 25\ \mu C$

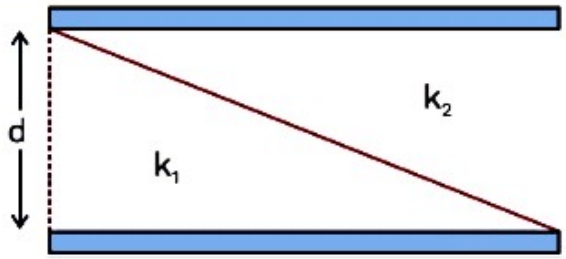
$y = 75\ \mu C$

The final situation is shown here,



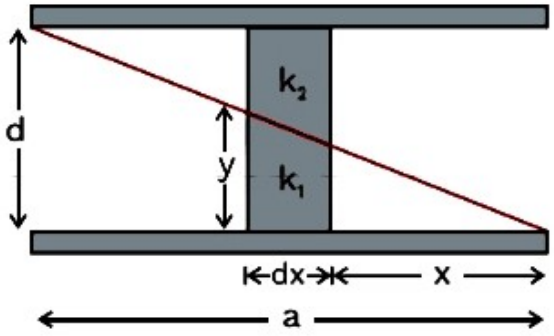
Important Problems based on Capacitor

Q.



There is a square plates of capacitor and the dimension is [a × a] and the separation between the plates is 'd'. Find the capacitance of such arrangement?

Sol.



Consider two infinitely small strips of width dx as shown.

$$\tan\theta = \frac{d}{a} = \frac{y}{x}$$

$$\frac{d}{a} = \frac{y}{x} \qquad \ldots(1)$$

$$dC_1 = \frac{adx\,\epsilon_0\,k_1}{y}$$

$$dC_2 = \frac{adx\,\epsilon_0\,k_2}{d-y}$$

As the combination of dC_1 and dC_2 is in series, so
Their net series combination

$$dc = \frac{dC_1\,dC_2}{dC_1 + dC_2}$$

$$= \frac{\frac{k_1 a dx\, \epsilon_0}{y} \times \frac{k_2 a dx\, \epsilon_0}{d-y}}{a dx\, \epsilon_0 \left[\frac{k_1}{y} + \frac{k_2}{d-y}\right]}$$

$$= \frac{k_1 k_2 a\, \epsilon_0\, dx}{k_1 d + (k_2 - k_1) y} \qquad \ldots(i)$$

As $x = y\frac{a}{d}$

$$dx = dy\left(\frac{a}{d}\right)$$

Replace dx to dy ...(ii)

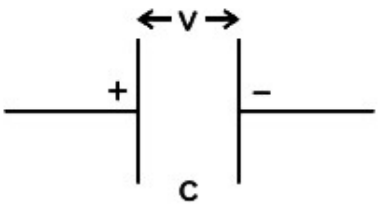
$$dC = \frac{k_1 k_2 a^2 \epsilon_0\, dy}{d[k_1 d + (k_2 - k_1) y]}$$

$$C = \int_0^d \frac{k_1 k_2 a^2 \epsilon_0}{d[k_1 d + (k_2 - k_1) y]} \cdot dy$$

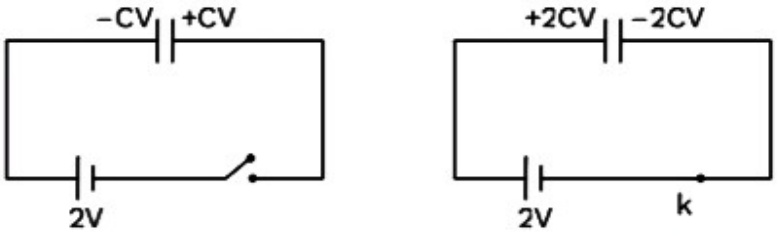
$$\boxed{= \frac{k_1 k_2 a^2 \epsilon_0}{(k_2 - k_1) d} \ln \frac{k_1 d + [k_2 - k_1] d}{k_1 d}}$$

Q. **There is a charged capacitance C, potential difference is V. If this capacitor is connected with a battery of potential difference 2V with opposite polarities, find out the loss of energy.**

Sol.



Initial energy stored in capacitor $E_1 = \frac{C}{2}V^2$. Now this arrangement is connected with battery of potential difference 2V, with opposite polarity.



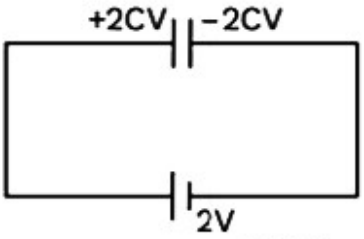
[Before key is closed] [After key is closed]

These figures show that the charge given by battery = 3CV

The work done by battery to provide this charge

$= [3CV]\,[2V] = 6CV^2$

Now



In the above figure, the energy stored is $= \frac{(2CV)^2}{2C} = 2CV^2$

$\therefore$ The Energy Loss

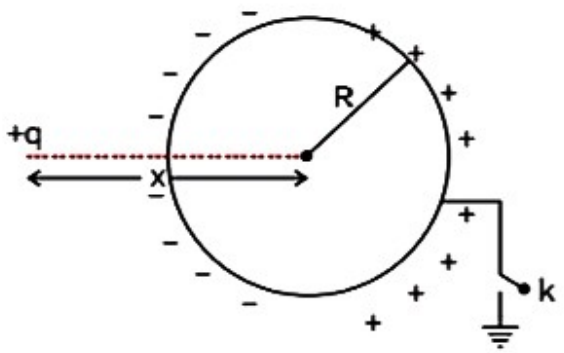
= Initial energy + W.D by battery − final energy stored in capacitor

$= \frac{CV^2}{2} + 6CV^2 - 2CV^2$

$= \frac{9}{2}CV^2$

Earthing of a Conductor

If any conducting device is connected to earth, then earth will make its electric potential equal to zero.



Let there is a sphere of radius R and before the key k is connected, the electric potential of sphere

$$V_{Net} = V_{+q} + V_{induced\ charge}$$

$$= \frac{kq}{x} + 0 \text{ [electric potential due to induced charge will be zero].}$$

Now this sphere is connected with earth, then suppose earth gives a charge q' to the sphere so that its potential should be equal to zero.
Now we have to calculate that charge

$$\frac{kq}{x} + \frac{kq'}{R} = 0$$

$$\frac{q'}{R} = -\frac{q}{x}$$

$$\boxed{q' = -q\frac{R}{x}}$$

If the outside charge moves either towards or away from the sphere, then a current will set up

$$\text{Current, } \frac{dq'}{dt} = +\frac{qR}{x^2}v$$

Q. **A capacitor of capacitance C_0 is charged to a potential V_0 and then isolated. A small capacitor C is then charged from C_0, discharged and then charged again. If the process is being repeated 'n' times, the potential of the large capacitor has now fallen to V. Calculate C.**

Sol. If an uncharged capacitor (C) is connected with charged capacitor [C_0], the common potential

$$V_1 = \frac{C_0V_0}{C + C_0}$$

Now this potential V_1 is gained by capacitor C and C_0, but 'C' is again discharged and again connected with C_0, then new common potential

$$V_2 = \frac{C_0V_1}{C + C_0} = \frac{C_0^2V_0}{(C + C_0)^2}$$

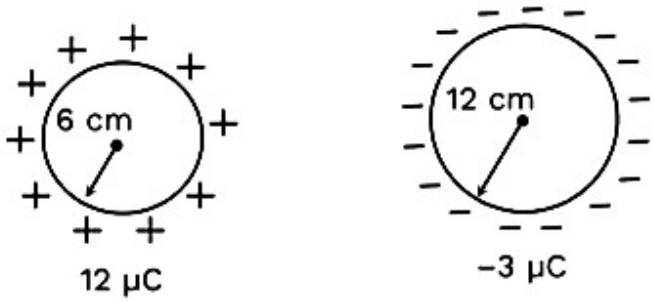
If the same process is repeated 'n" times

$$V_n = V = \left[\frac{C_0}{C + C_0}\right]^n V_0$$

$$C = \left[\left(\frac{V_0}{V}\right)^{1/n} - 1\right]C_0$$

Q. **Two isolated spherical conductors have radii 6 cm and 12 cm respectively. They have charges of 12 μC and -3μC. Find the charges after they are connected by a conducting wire. Also find the common potential after redistribution?**

Sol.

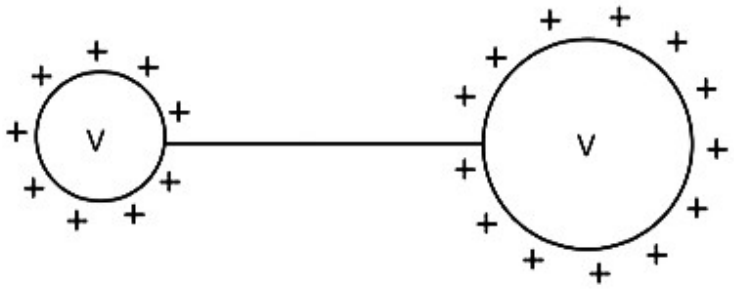


After connecting through the conducting wire charge will be flowing from higher potential to lower potential, till both are not attaining a common potential

Net charge = (12 – 3) μC = 9μC

As both spheres gain common potential, then charge on each sphere is proportional to their radius

$$q_1 = \frac{R_1}{R_1 + R_2} q_{Total}$$



$$= \frac{6}{18} \times 9\mu C = 3\mu C$$

$$q_2 = \frac{12}{18} \times 9\mu C = 6\mu C$$

The common potential

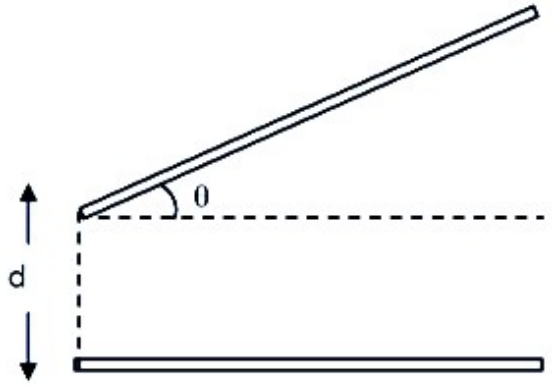
$$V_c = \frac{\text{Total Charge}}{\text{Total Capacitance}}$$

$$= \frac{9 \times 10^{-6}}{4\pi \epsilon_0 [R_1 + R_2]}$$

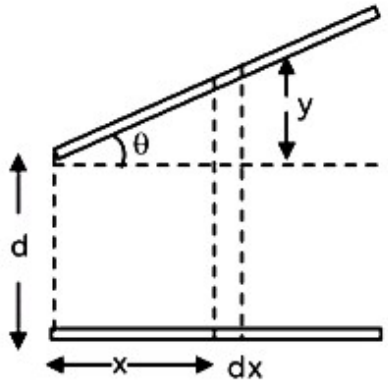
$$= 4.5 \times 10^5 V$$

Q. **A capacitor has square plates each of side 'a' making an angle θ with each other as shown in figure. Show that for small θ, the capacitance is given by**

$$C = \frac{a^2 \epsilon_0}{d}\left(1 - \frac{\theta a}{2d}\right)$$



Sol.



If we take a small element of width dx at a distance of x from the left end, then its capacitance will be

$$dC = \frac{a dx \epsilon_0}{d + y}$$

$$y = x \tan \theta$$

$$\therefore \quad dC = \frac{a \epsilon_0 \, dx}{d + x \tan \theta}$$

As is θ small

$\tan\theta \approx \theta$

$$dC = \frac{a\,\epsilon_0\,dx}{d + x\theta}$$

$$= \frac{a\,\epsilon_0\,dx}{d\left(1 + \frac{x\theta}{d}\right)} = \frac{a\,\epsilon_0}{d}\left(1 + \frac{x\theta}{d}\right)^{-1}$$

$$C = \frac{a\,\epsilon_0}{d}\int_0^a \left(1 - \frac{x\theta}{d}\right)dx$$

$$= \frac{a\,\epsilon_0}{d}\left[x - \frac{x^2\theta}{2d}\right]_0^a$$

$$= \frac{a^2\,\epsilon_0}{d}\left[1 - \frac{a\theta}{2d}\right]$$